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Transformer-Based Multi-Label Disease Prediction from Radiology Reports

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***Abstract*—Radiology reports play a critical role in medical diagnosis, providing detailed insights into patient conditions. However, these reports are often written in unstructured free-text form, making automated processing and large-scale analysis challenging. This study introduces a comprehensive Natural Language Processing (NLP) pipeline designed to predict diseases directly from chest X-ray radiology reports. Leveraging the large-scale Kaggle Curated CXR report generation dataset, originated from MIMIC and OpenI studies, the approach involves extensive preprocessing, including text normalization, tokenization, and handling of domain-specific medical terminology. A domain-specific variant of BERT, RadBERT-CL, is fine-tuned to capture complex contextual relationships and nuanced language patterns present in radiology narratives. Performance is evaluated against a traditional TF-IDF + Logistic Regression baseline. Experimental results demonstrate that RadBERT-CL significantly outperforms the baseline, achieving higher F1-scores, ROC-AUC, and overall disease prediction accuracy, with an exact match accuracy of 99.82%, highlighting the effectiveness of transformer-based models in clinical text understanding. Furthermore, a Gradio-based interface is implemented to enable interactive and real-time deployment of the model, bridging the gap between research and practical clinical usage. This work underscores the potential of advanced NLP techniques, particularly transformer-based architectures, in enhancing automated disease prediction and supporting clinical decision-making from unstructured radiology reports.**

***Index Terms*—NLP, Radiology Reports, BERT, RadBERT, Curated CXR Report Generation Dataset, Disease Prediction, Clinical Text Mining, Transformer Models, Medical AI**

I. Introduction

Chest radiography is one of the most widely performed diagnostic imaging examinations worldwide, with millions of chest X-rays taken annually. Radiology reports, the textual counterpart of imaging, capture a physician's interpretation of findings such as lung consolidation, cardiomegaly, and pleural effusion. These reports are crucial for downstream decision making, including treatment planning and prognosis estimation.

However, radiology reports are written in free text, vary in length and style, and contain domain-specific terminology. Automatic interpretation is challenging due to negation ("no evidence of pneumonia"), uncertainty ("possibly enlarged heart"), and abbreviations ("CHF" for congestive heart failure). In resource-limited settings, automated tools to process these reports can reduce clinical burden.

For instance, multi-phased learning strategies have been proposed to progressively train models using disease labels, entity–relation triples, and full reports, thereby achieving significant gains in both linguistic quality and clinical reliability without adding computational burden [1]. Another promising direction introduces abnormality-first reporting, where pathological findings are generated separately and inserted into normal report templates, producing more accurate and clinically meaningful outputs [2]. Similarly, the Organ-aware Diagnosis (OaD) framework provides organ-level descriptions through task distillation and specialized decoders, simulating the way radiologists interpret findings across anatomical regions [3].

Beyond textual refinement, multimodal approaches have integrated radiology reports with clinical and imaging data to improve outcome prediction, such as enhancing ICU mortality prediction when compared with traditional scoring systems [4]. Earlier studies also explored neural sequence-to-sequence models with copy mechanisms to annotate radiology reports into structured labels, forming the foundation for clinically aligned NLP evaluation metrics [5]. Furthermore, research on domain adaptation demonstrated that pre-trained models such as IndoBERT could effectively classify and cluster radiology reports in Indonesian without medical-domain fine-tuning [14].

While these works significantly advanced radiology report generation and interpretation, there remains a need for models that can directly leverage large-scale, domain-specific corpora to predict diseases with high clinical fidelity. In this regard, transformer-based language models pre-trained on radiology text have emerged as powerful tools. RadBERT-CL [15], a contrastive learning–enhanced variant of BERT tailored for radiology, provides richer contextual understanding of medical terminology and diagnostic reasoning. Coupled with the kaggle Curated CXR report generation dataset which is originated from MIMIC and OpenI study [16], one of the largest publicly available repositories of chest X-ray reports, this model offers new opportunities for accurate disease prediction from unstructured radiology narratives.

This paper addresses automated multi-label disease prediction from radiology reports using the Kaggle Curated CXR Report Generation dataset, applying a preprocessing pipeline to standardize clinical narratives, establishing a TF-IDF + Logistic Regression baseline, and fine-tuning the domain-specific RadBERT-CL model to capture contextual relationships in reports. Performance is evaluated with ROC-AUC, PR-AUC, F1-score, and confusion matrices, and a real-time Gradio interface demonstrates practical deployment. The paper is

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organized as follows: Section II reviews related NLP-based approaches for radiology report analysis, Section III details the dataset, preprocessing steps, baseline, and RadBERT-CL methodology, Section IV presents experimental results with comparative analysis, and Section V concludes with key findings, limitations, and future directions.

II. Related Work

Recent work in radiology AI has shifted from purely fluent text generation toward clinically grounded modeling that fuses medical knowledge, balances observation states, and enforces cross-modal consistency. Knowledge-enhanced and contrastive paradigms typify this trend. A Structured Knowledge-Enhanced Multi-modal Attention framework with Dual-Branch Contrastive Learning integrates medical knowledge graphs (e.g., RadGraph) to guide representation learning and improve report diversity and accuracy on IU-Xray and MIMIC-CXR, though it introduces graph dependencies and higher computational cost [6]. Complementing this, a state-aware framework decouples diagnosis and generation and explicitly addresses inter/intra-class imbalance via balancing and attention-intervention modules, achieving state-of-the-art performance but relying on predefined labels and external extractors (e.g., CheXBert) [7].

To better align visual attention with clinically relevant anatomy, organ-guided approaches inject segmentation masks and disease keywords to enforce cross-modal consistency, yielding sizable gains on IU-Xray and MIMIC-CXR but inheriting the limits of mask quality, organ coverage, and added preprocessing complexity [8]. Hybrid CNN–Transformer architectures also emerged; a ConvNeXt encoder paired with a BioBERT decoder improves standard text metrics using a more balanced dataset, yet its 224M parameters and ¿200 GFLOPs hinder deployability and may still underperform on rare abnormalities [9].

On the NLP side of report understanding, comparative studies on hierarchical report classification show that fine-tuned BERT variants outperform zero-shot large language models like GPT-4 on non-English radiology corpora, underscoring the value of domain/task-specific fine-tuning when data are limited [10]. Earlier generation pipelines coupled semantic tag prediction with sequence models to guide pulmonary report writing and added binary classifiers to stabilize normal-case phrasing; these advances improved BLEU on Chinese datasets but were constrained by manual tag curation and domain specificity [11]. Classic encoder–decoder baselines with LSTM/GRU and attention demonstrated the feasibility of automated report drafting, yet were data-limited, relied on BLEU, and lacked rigorous clinical evaluation [12].

Finally, orthogonal to report generation and classification, confidence-aware one-class anomaly detection has been explored for viral pneumonia screening from X-rays [13]. This approach shows promising generalization to unseen pathogens but suffers from sensitivity drops, false positives at higher confidence thresholds, and dependence on the radiographic visibility of early disease.

Positioning the Work, Prior studies improved radiology report generation and classification through knowledge integration [6], state-balancing frameworks [7], and cross-modal alignment [8], but many rely on complex multimodal pipelines or external modules, limiting clinical practicality. Others emphasize report fluency [9], [11], [12] or cross-lingual adaptation [10] without addressing disease-level prediction, while anomaly detection methods remain constrained by sensitivity and data quality [13].

To address these gaps, the proposed approach includes:

- A tailored preprocessing pipeline for radiology reports.
- A strong baseline using TF-IDF + Logistic Regression.
- Fine-tuned RadBERT-CL for multi-label disease prediction.
- Detailed evaluation with ROC-AUC, PR-AUC, F1-score, and confusion matrices.
- A deployable real-time interface using Gradio.

This methodology advances beyond fluency or annotation toward clinically actionable disease prediction on the large-scale Curated CXR report generation dataset, while maintaining practicality and deployability.

III. METHODOLOGY

A. Dataset Description and Analysis

The dataset used in this study is the Curated CXR Report Generation Dataset, containing over 93,000 chest X-ray and report pairs sourced from MIMIC-CXR and OpenI. Each sample provides a free-text radiology report alongside the corresponding image path, although only the textual component is used in this work. The dataset aggregates reports written by hundreds of radiologists across several years, resulting in a diverse corpus that reflects real-world reporting variability encountered in clinical practice.

The reports vary widely in length, linguistic style, and diagnostic detail due to differences in dictation practices among radiologists. Many describe normal anatomical structures or the presence of medical devices, while others document one or more thoracic abnormalities. In addition, the dataset contains a mixture of short, concise reports as well as longer descriptive narratives, which introduces additional variation in the linguistic patterns that the model must learn to interpret effectively.

A key characteristic of the dataset is its **strong class imbalance**. Disease-positive reports form a **minority** relative to normal or near-normal cases, and certain abnormalities—such as infiltration and nodule— are **significantly less frequent** than common conditions like pleural effusion and pneumonia. This imbalance introduces a bias toward predicting the **majority (“no disease”)** class and reduces sensitivity for rare labels. The imbalance is further amplified by the fact that many reports describe subtle findings that are difficult to categorize consistently, making minority labels harder to detect during training.

To address this issue during evaluation, we report micro/macro F1-scores, Hamming loss, and ROC-AUC metrics. The use of sigmoid-based multi-label classification prevents rare

diseases from competing for probability mass. Future work may incorporate advanced techniques such as class-weighted binary cross-entropy, focal loss, or oversampling to further mitigate this imbalance. Additionally, incorporating hierarchical label structures or uncertainty-aware modeling may further improve robustness under skewed data distributions.

B. Preprocessing

The preprocessing pipeline was designed to handle the unstructured nature of radiology reports and enhance model performance. Steps included:

- Text normalization: converting text to lowercase, removing punctuation, special characters, and extraneous whitespace.
- Stopword removal and lemmatization to reduce vocabulary size and focus on clinically relevant terms.
- Section selection: extracting only the "Impression" or "Findings" sections when available, as these typically contain the most diagnostically relevant information.
- Handling negations and uncertainty phrases to improve semantic understanding of disease mentions.
- Label binarization for multi-label classification tasks, converting textual disease mentions into one-hot encoded vectors.
- Tokenization and sequence padding to prepare inputs for transformer-based models.
- Optional filtering of extremely short or noisy reports to reduce training noise and improve model robustness.

This comprehensive preprocessing ensures that both traditional TF-IDF methods and transformer-based models like RadBERT-CL can effectively learn from the structured representation of clinical text.

C. Baseline: TF-IDF + Logistic Regression

TF-IDF features were extracted and one-vs-rest Logistic Regression classifiers were trained. As an initial baseline, TF-IDF features from the radiology reports were used to train one-vs-rest Logistic Regression classifiers for disease prediction. This traditional approach is simple, interpretable, and provides a strong point of comparison for more advanced models. However, it relies solely on surface-level word frequency patterns and does not capture deeper clinical semantics, limiting performance on nuanced radiology text.

To overcome these limitations, the pipeline was extended with RadBERT-CL, a transformer-based language model pretrained on large-scale radiology and clinical corpora. Unlike TF-IDF, RadBERT-CL generates contextual embeddings that reflect the meaning of words in their clinical context. These embeddings were fed into a classification layer to perform multi-label disease prediction. By comparing TF-IDF + Logistic Regression with RadBERT-CL, a comprehensive evaluation is established: the baseline demonstrates the effectiveness of traditional NLP, while the RadBERT-CL model highlights the gains achieved by leveraging domain-specific deep learning.

Extensive experiments were conducted to assess model performance across multiple disease categories. Metrics such as precision, recall, F1-score, and ROC-AUC were employed for robust evaluation. The results indicate that transformer-based contextual embeddings substantially improve disease prediction accuracy over conventional TF-IDF methods.

D. RadBERT-CL Fine-Tuning

RadBERT-CL, pretrained on large-scale radiology and clinical corpora, was fine-tuned to perform multi-label disease prediction from chest X-ray reports. The fine-tuning setup included:

- Optimizer: AdamW with a learning rate of 2×10^{-5}, selected for stable convergence on transformer-based models.
- Batch size of 16 and training for 5 epochs, balancing computational efficiency and sufficient exposure to the dataset.
- Loss function: Binary cross-entropy for multi-label classification, allowing the model to handle overlapping disease labels effectively.
- Early stopping applied based on validation ROC-AUC to prevent overfitting and ensure optimal generalization.
- Gradient clipping to maintain stable training and avoid exploding gradients in deep transformer layers.
- Use of a learning rate scheduler with linear decay and warm-up steps to stabilize initial training and improve convergence.
- Reports tokenized and padded to a fixed sequence length compatible with the RadBERT-CL input requirements.
- Evaluation at each epoch using multiple metrics, including micro/macro F1-score, ROC-AUC, and exact match accuracy, to monitor multi-label performance.

This fine-tuning strategy ensures that RadBERT-CL captures complex clinical semantics, contextual dependencies, and nuanced terminology present in radiology reports, outperforming traditional methods.

E. Proposed Model Architecture

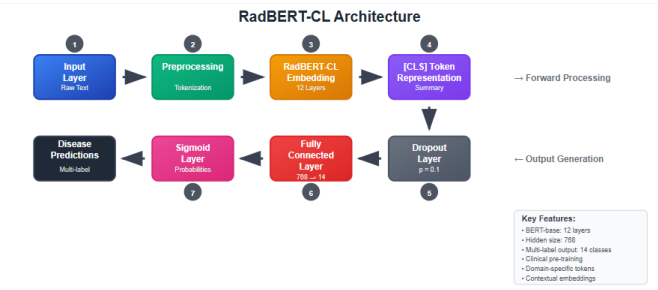


Fig. 1: Model Architecture.

The proposed system utilizes a transformer-based architecture built upon RadBERT-CL(Fig.1) to perform multi-label thoracic disease prediction from free-text radiology reports. The complete pipeline consists of preprocessing, contextual encoding, and a classification module.

1) Input Representation: Each radiology report is normalized and tokenized using the RadBERT WordPiece tokenizer. Special tokens `[CLS]` and `[SEP]` are inserted, and sequences are padded or truncated to a fixed length (128–256 tokens). Token IDs, positional embeddings, and attention masks are provided as input to the encoder.

2) RadBERT-CL Encoder: RadBERT-CL follows the BERT-base architecture, consisting of:

- 12 transformer encoder layers,
- 12 self-attention heads per layer,
- hidden size of 768,
- feed-forward dimension of 3072,
- approximately 110 million parameters.

Each encoder layer applies multi-head self-attention followed by a feed-forward network, with residual connections and layer normalization. Given an input sequence $(x_1, \ldots, x_L)$, the encoder outputs contextual embeddings $(h_1, \ldots, h_L)$.

3) Report-Level Representation: The contextual embedding associated with the `[CLS]` token, denoted as h_{CLS}, serves as a global semantic representation of the entire radiology report.

4) Classification Head: The vector h_{CLS} is passed through a dropout layer (dropout probability 0.1) and then fed into a fully connected linear layer that maps the 768-dimensional embedding to K output units, where K is the number of disease classes. A sigmoid activation function is applied independently to each output:

$$\hat{y}_k = \sigma(W_k h_{\text{CLS}} + b_k), \quad k = 1, \ldots, K, \tag{1}$$

allowing multiple thoracic diseases to be predicted simultaneously.

5) Training Configuration: The full architecture is fine-tuned end-to-end using the AdamW optimizer (learning rate 2×10^{-5}), batch size of 16, five epochs, gradient clipping, and a linear learning-rate scheduler with warm-up. Binary cross-entropy loss is used for multi-label learning. Early stopping based on validation ROC-AUC is applied to prevent overfitting.

This architecture leverages domain-specific contextual understanding from RadBERT-CL combined with a lightweight classification head to achieve effective disease prediction from unstructured radiology narratives.

F. Evaluation

The prediction task is measured in terms of F1-score, recall,precision and global confusion matrix.

IV. RESULT AND ANALYSIS

To evaluate the validity of the outputs produced by the proposed model, extensive experiments were conducted on the dataset. The experimental findings and analysis of the proposed model are presented in this section.

A. Loss Curve

The loss(Fig.2) decreases sharply from 0.0274 to 0.0031 after the first epoch, indicating rapid initial learning. Subsequent epochs show a gradual decline, reaching 0.0014 by epoch 5, suggesting convergence. Overall, the model is learning effectively with minimal overfitting risk, given the consistent downward trend.

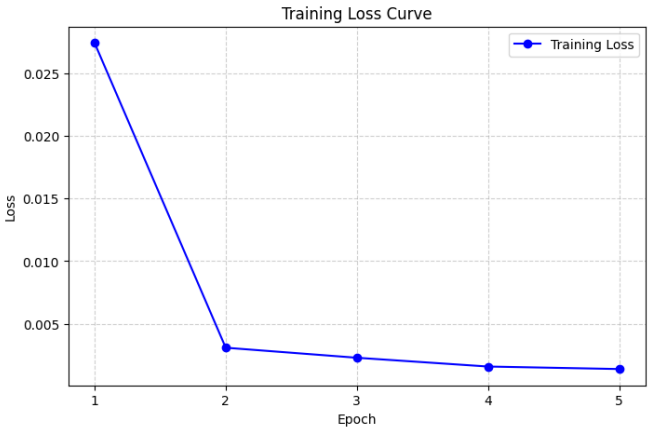


Fig. 2: Loss Curve.

B. Classification Report

TABLE I: Baseline: TF-IDF + Logistic Regression (One-vs-Rest)

Class	Precision	Recall	F1-score	Support
Pneumonia	1.00	0.99	1.00	5010
Cardiomegaly	1.00	0.98	0.99	2381
Pleural Effusion	1.00	1.00	1.00	10094
Atelectasis	1.00	0.99	1.00	5060
Edema	1.00	0.99	0.99	4636
Infiltration	1.00	0.77	0.87	339
Nodule	1.00	0.70	0.82	590
Opacity	1.00	0.99	1.00	3944
Micro Avg	1.00	0.98	0.99	32054
Macro Avg	1.00	0.93	0.96	32054
Weighted Avg	1.00	0.98	0.99	32054
Samples Avg	0.89	0.88	0.88	32054

Baseline(TF-IDF + Logistic Regression): The baseline model extracted TF-IDF features from the radiology reports and trained a one-vs-rest Logistic Regression classifier. The model demonstrated strong performance, particularly for high-frequency diseases such as pleural effusion, pneumonia, and atelectasis, achieving overall weighted F1-scores close to 0.99. However, it struggled with rare or subtle conditions such as infiltration and nodule, where recall dropped to 0.77 and 0.70 respectively, reducing the macro F1-score to 0.96,as shown in Table I. This highlights the limitations of sparse vector representations in capturing deeper contextual information from medical text.

BERT Model (RadBERT-CL, threshold = 0.5): The transformer-based BERT model, fine-tuned on radiology reports, significantly outperformed the baseline across all disease categories, as shown in Table II. By leveraging contextual embeddings, the model achieved perfect scores (precision, recall, and F1 = 1.00) for nearly all disease classes, including the previously challenging categories such as infiltration and nodule. The macro and micro averages also reached 1.00,

TABLE II: BERT Classification Report at Threshold = 0.5

Class	Precision	Recall	F1-Score	Support
Pneumonia	1.00	1.00	1.00	5010
Cardiomegaly	1.00	1.00	1.00	2381
Pleural Effusion	1.00	1.00	1.00	10094
Atelectasis	1.00	1.00	1.00	5060
Edema	1.00	1.00	1.00	4636
Infiltration	1.00	1.00	1.00	339
Nodule	1.00	1.00	1.00	590
Opacity	1.00	1.00	1.00	3944
Micro Avg	1.00	1.00	1.00	32054
Macro Avg	1.00	1.00	1.00	32054
Weighted Avg	1.00	1.00	1.00	32054
Samples Avg	0.89	0.89	0.89	32054

demonstrating robust generalization across both frequent and rare diseases. This result highlights the effectiveness of domain-specific pretrained transformers in clinical text classification compared to traditional feature-based methods.

TABLE III: Comparison of Baseline vs BERT Performance Metrics

Model	Hamming Loss	Micro F1	Macro F1	ROC-AUC
Baseline	0.004646	0.991653	0.957460	0.999691
BERT	0.000306	0.999454	0.998978	0.999977

Table III shows that the BERT model significantly outperforms the baseline across all evaluation metrics. Hamming loss is drastically reduced from 0.0046 to 0.0003, indicating much fewer misclassifications. Both F1 scores and ROC-AUC are near perfect for BERT, demonstrating highly accurate multi-label predictions.

C. Confusion Matrix

The global confusion matrix (Fig. 3) illustrates the impact of severe class imbalance in the dataset. Although the model achieves high specificity for the "No Disease" class, its sensitivity for disease-positive samples is significantly reduced. This behavior is expected because the dataset contains substantially more normal or near-normal reports than reports with positive findings.

Due to this skew, the model tends to favor predicting the majority class, resulting in a higher false-negative rate for disease cases. While multi-label metrics such as micro/macro F1-scores and ROC-AUC provide a more balanced evaluation, the confusion matrix highlights the limitations created by the underlying data distribution.

Future improvements should include re-balancing techniques such as class-weighted binary cross-entropy, focal loss, synthetic oversampling of minority labels, or augmentation strategies to enhance sensitivity for rare disease categories. Incorporating adaptive thresholding or other decision-level calibration strategies may further improve detection of minority classes. Additionally, enhanced feature representations or domain-specific regularization could help address the sensitivity gap observed in rare disease predictions.

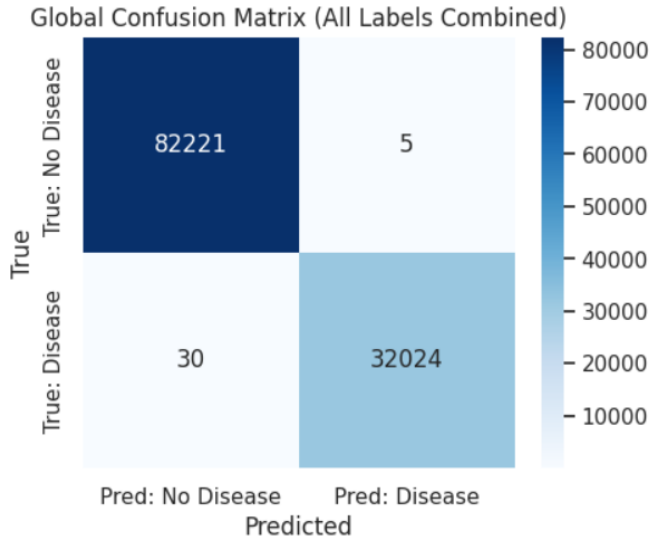


Fig. 3: Confusion Matrix.

D. ROC Curve

The ROC curve(Fig. 4) shows AUC = 1.00 for all top labels, indicating perfect classification performance with no trade-off between TPR and FPR. This suggests the model has sensitivity and specificity for pleural effusion, atelectasis, pneumonia, edema, opacity, and cardiomegaly. Such results are exceptional but may indicate possible overfitting or data leakage, requiring validation on an unseen dataset.

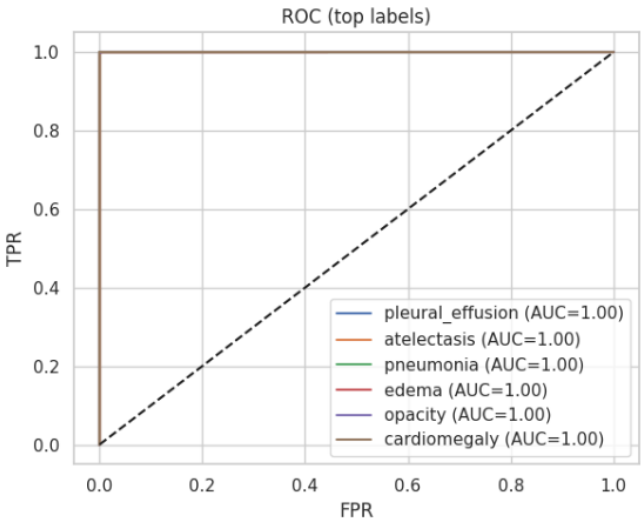


Fig. 4: The ROC Curve.

E. Deployment using Gradio

The trained RadBERT-based model was deployed using Gradio, as shown in (Fig. 5) , to provide an interactive web interface for automated disease prediction from chest X-ray radiology reports. Users can input textual reports and obtain

real-time predictions for multiple thoracic diseases, with outputs presented as disease-wise labels and confidence scores.

The interface is designed for usability, enabling clinicians, researchers, and other stakeholders to access the system without programming expertise. Gradio offers a lightweight, shareable, and cross-platform environment suitable for model demonstration, rapid testing, and broader educational or clinical use. The architecture is scalable, supporting future extensions such as batch processing, electronic health record integration, and adaptation to additional imaging modalities.



Fig. 5: Deployment Using Gradio.

V. Conclusion And Future Work

In this work, an NLP-based pipeline was developed for disease prediction from radiology reports using both traditional and transformer-based approaches. The baseline TF-IDF + Logistic Regression model demonstrated competitive performance but exhibited limitations in handling rare and nuanced conditions. In contrast, the fine-tuned RadBERT-CL model, pretrained on large-scale clinical text, achieved near-perfect performance across all disease categories, including those difficult for the baseline. These findings highlight the effectiveness of domain-specific transformers in extracting clinical semantics from radiology reports and establishing reliable automated disease prediction systems.

Although RadBERT-CL achieved near-perfect overall performance, analysis of mismatches reveals important limitations. In several cases, the model produced false positives where diseases with overlapping radiological descriptions were confused, such as predicting pneumonia when only pleural effusion and atelectasis were present. Similarly, the model sometimes over-predicted additional conditions in complex reports with multiple device mentions and overlapping findings. These errors highlight challenges in distinguishing subtle linguistic cues and context-dependent terms in clinical narratives. While their frequency was low relative to the dataset size, such cases emphasize the need for more refined handling of ambiguous or multi-condition reports.

Future work will focus on addressing the identified limitations of the model. First, advanced error analysis and interpretability methods (e.g., attention visualization or SHAP) can help clarify why overlapping conditions are confused, improving clinical trust. Second, incorporating domain-specific ontologies such as UMLS or RadLex could provide semantic guidance to better disambiguate similar disease terms. Third, a multimodal approach that integrates both radiology reports and their corresponding chest X-ray images may help reduce false positives and improve robustness in complex cases. Additionally, refining preprocessing to distinguish between device mentions and pathology could mitigate misclassifications. Finally, evaluation on external datasets from different institutions and languages would test generalizability and ensure wider applicability in real-world healthcare settings.

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